Multimodal Dataset of Welding Process Images and Power Source Data for Deep Learning

Tommi Aihkisalo
VTT Technical Research Centre of Finland
Oulu, Finland
tommi.aihkisalo@vtt.fi

Petri Tikka
VTT Technical Research Centre of Finland
Tampere, Finland
petri.tikka@vtt.fi

***Abstract*— This paper presents a multimodal KemppiPowerImages dataset named after the data contributors. This dataset was captured during welding trials, containing power source time series data synchronized with imagery. This novel and unique dataset enables training of deep learning algorithms combining machine vision and time series processing. Potential tasks include generative tasks and anomaly detection for quality monitoring and process control applications and services for manufacturing industry utilizing welding. The dataset comprises welding current and voltage values and over 34,000 images, aligned and synchronized. We describe the capture process and utilized welding parameters, and processing steps required to reach the final dataset. We also assess data quality and illustrate one potential deep learning application achievable with this dataset.**

Keywords—dataset, deep learning, multimodality, manufacturing, welding

I. Introduction

This paper presents a novel and unique dataset collected from several welding test runs and its initial processing steps. We named the dataset after the main contributors as KemppiPowerImages dataset [1] or KPI for short. The novelty, not seen in any other reviewed datasets, includes the two simultaneously collected modalities; the video image frames, and power source data synchronously collected during the welding runs. The welding process was fillet welding [2] with Gas Metal Arc Welding, GMAW, method on S335MC structural steel sample structures. The data capture processes for the image and power source data were concurrent on independent data acquisition equipment for both. Despite the concurrency, the mentioned modalities were captured in their own distinctive time domains. That distinctiveness left us with a problem of two different time domains which required alignment and synchronization of both modalities together.

The alignment and synchronization procedure produced a coherent and concurrent timed dataset including timed pairs of the video frame images and power source time series data consisting of the current and voltage. The time domain synchronization between both modalities is based on the visual inspection of the video frames to identify the ignition of the welding arc which can be identified reliably also from the welding current and voltage data. This allows us to align reliably the whole duration of the welding process from arc ignition to stop for both modalities. Outside of the start and stop the alignment does not apply due to an unmarked manual control resulting in different runtimes of the data captures. The further processing included dividing the aligned video image frames equally to the whole duration of the aligned time series data.

The whole processing resulted in lengths of power source time series paired and indexed with the video frames. The dataset can be repurposed for utilizing only a single modality or all of them depending on the application. The conceptual contents of the final dataset of the welding runs' power source data synchronized with the individual video image frames are presented in Fig 1.

II. Related Work

Many well enough documented related datasets exist in the field. These are datasets for certain manufacturing and quality control purposes but most lacking the multimodality in the data itself e.g., according to [3]. Many of the datasets published in the manufacturing are usually either numerical values or in some fewer cases images only. One welding and process monitoring related dataset is [4] collected for multivariate evaluation and study in [5] for the impact of six different welding parameters on the achieved welding depth and other geometrical dimensions in steel-copper lap joints in laser welding process.

Usually, image datasets are meant for defect or other varying anomalous factor detection with AI-ML assisted machine vision tasks, e.g., object detection or segmentation. These usually have additional annotation data for the images on the side for supervised training methods. One example from the manufacturing domain and quality control aspect is [6] containing images from the steel manufacturing and annotations for machine vision segmentation and classification.

For truly multimodal datasets one must look other than the manufacturing and welding domain. Such as CUB dataset [7] contains images of birds and annotations containing a description of the image contents in natural language suited e.g., for generative machine learning usage. However, some purposeful aggregation of the several existing single modality datasets can be done. Provided of course that the represented domains in the modalities are cojoined and the connecting factor

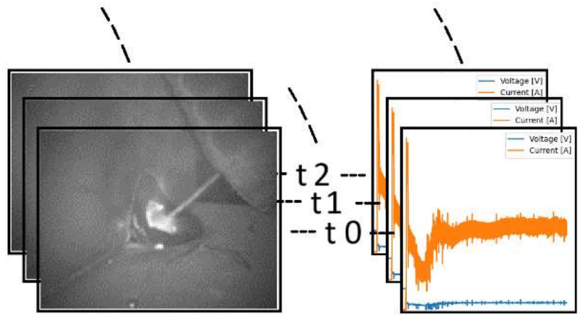

Fig. 1. Conceptual content of the final dataset with aligned and synchronized power source time series and the video frames indexed incrementally starting from t0.

is identifiable. One such example is presented in [8] and in [9], mainly discussing the aspects of the multi-modal models. They are using combined two datasets, well known MNIST and street-view house number SVHN, as an example of the multimodal dataset for demonstration.

The review of the related work and datasets suggests that this KPI dataset is novel and unique in its multimodality in the manufacturing domain. KPI dataset consists of temporally synchronized and connected images and the time series that introduce new possibilities for the machine learning purposes especially in the manufacturing and welding domain.

III. Data Acquistion Runs

The dataset is a collection of image frames from captured video feeds and measurements of power source voltage and current as time series. Data was collected from several test runs, each utilizing both data acquisition systems. However, not all test runs yielded fully usable data due to errors such as insufficient or inconsistent data capture runtimes, leading to the exclusion of some runs. This leaves us with 16 full data collections, in where the very first test run was merely an experimental producing data with different parameters than the subsequent runs. Data from the very first run is included as “deemed useful”.

All the welding test runs were on S335MC steel material with filler wire diameter of 1.2 mm producing joint type fillet welds with 5 mm of throat thickness. The test welding setup used Kemppi AX MIG Welder 500 Pulse+ -welding machine with a Yaskawa robot, welding imaging camera and customized power source data acquisition equipment. The shielding gas was Argon + 18 % CO_2 mixture. The test welding setup with the test structure, welding torch on a robot arm and a camera setup is illustrated in Fig 2. The welding parameters, or the Welding Procedure Specifications (WPS), regarding the set welding currents, voltages, position among others are listed on Table 1 in the appendix.

Fig. 2. Test welding run setup with steel structure for fillet welding, active welding torch, camera equipment and some non-disclosed additional equipment.

IV. Data Acquistion Results

The completed test runs, with the resulting data and their respective dimensions detailed in Table 3 in the appendix. Only the datasets deemed to cover full and complete coverage in both modalities are included to allow proper alignment and synchronization of both modalities. Variations in the data acquisition processes and resulting synchronization are reflected in the varying maximum time series length that can be fit between the images. The visually identified welding arc start and end times for the images and power source time series are included in the table. The timing for the images is read from the image file names enclosing the image capturing time stamp. In the time series each data point is timed with the help of known sampling rate of 100 kHz or 80 kHz starting from zero increasing sequentially and linearly until the very last captured data sample.

Each time series were trimmed to the shortest existing time series in the test run. The initial run was conducted with different sampling rates and settings for trial purposes to fine-tune data acquisition procedures. The image frames are grayscale images of 720 x 540 or 1440 x 1080 pixels in the initial and successive data acquisition runs respectively. For the final dataset a 300 x 300 pixel Region of Interest (RoI) was separated from them.

A. *Time Resolution*

The timestamps originate from two independent data acquisition systems: the camera system and the power source data capturing equipment. The image frames were timestamped and time recorded on their file names while timestamps for the power source data were reconstructed using the known sampling rate. To synchronize the timelines from the two separate clock domains, initial time domain alignment was necessary. Before the alignment we consider the time resolution and other data dimensions in each of the data modalities.

- Image Data: Each image frame is individually timestamped, with date and time encoded in the file name. The timestamps are provided in a 24-hour format with a resolution up to the microsecond level. The mean time difference between consecutive image captures is approximately 33.35 milliseconds, equating to around 30 frames per second. On average, this allows for 1,333 to 2,666 power source data points between each image frame captured depending on the power source sampling rates.
- Power Source Data: Lacking individual timestamps, the data consists of a known number of samples with a sampling rate of 80,000 samples per second, resulting in a time resolution of 12.5 microseconds. 100,000 samples per second would mean resolution of 10.0 microseconds.

Aligning the timestamps for the image frames required converting and rescaling the datetime stamps to a simpler ascending microsecond format for consecutive frames. Synthetic timestamps were generated for each power source data point based on the known sampling rate. This process resulted in two independent timelines that required synchronization through well identifiable and time events present in both data modalities. Such an identified event would allow setting the same aligned zero time for all the modalities.

B. *Time Synchronization*

The start and end of the welding process, visually identifiable in the images by the ignition and dying down of the welding arc, respectively, serve as key events for synchronization. These events are also detectable in the electrical power source data. Each test run begins with the ignition of the welding arc shortly after data capture starts and ends sometime after the arc ceases. By recording the start and stop times of the welding arc in both the images and the electrical power source data, synchronization of the timelines is achieved.

The arc ignition is indicated by a rising edge of the welding current level and a simultaneous voltage drop, as shown in Fig 3(a) where current peaks to over 700 amps and voltage drops closer to zero. Conversely, a drop in the current and voltage marks the end of the welding process as illustrated in Fig 3(b) where the welding current drops to zero with the voltage. Visual inspection of the captured image frames allows identification of the timestamps where the welding arc first and last appears. This determines the start and stop limits of the welding run time sequence in the image capture time domain, Fig 4.

Identifying the start and stop times of the welding arc in both time domains allows for precise alignment. By zeroing and aligning the start and end times to match spreads the images evenly along the power source data timeline. Given the image sampling rate of approximately 30 frames per second, there are several power source data points between each pair of consecutive images, in practice approximately 1,333 to 2,666 as determined earlier. This method aligns the two unconnected time domains present in the test runs.

Higher-level delays in data collection equipment, such as those caused by transmission, buffering, and timestamping, are irrelevant if reasonably steady as only the rising or falling edge of the power and emergent visibility of the welding arc are determinable for this alignment. As a note for future applications, the camera supplier reports an average delay of 46.2 milliseconds from the image cell to the fully timestamped file, and about 0.31104 milliseconds from the camera cell to memory.

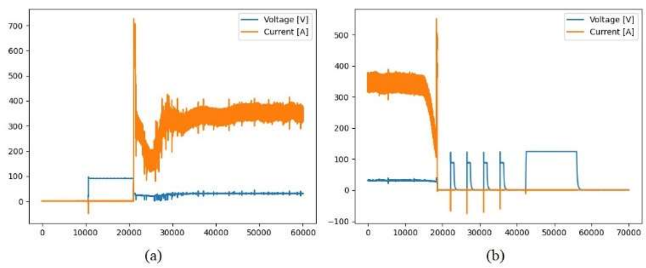

Fig. 3. Welding current and voltage. (a) arc ignition and the start of welding, indicated by a jump in current and a drop in voltage after timestep number 20,000 in the power source data. (b) Arc stall and the end of welding, indicated by a drop in current and

The very first image according to the information provided by the camera manufacturer gets delayed at its worst by 0.31104 milliseconds meaning around 24 data points at 80 kHz sampling rate for the power source data capturing. This is provided that the image capture event starts the same time as the power source data shows rising edge in the current values. The exact moment could be determined better if a high-speed camera were used, but it is not essential here.

The source of uncertainty in the timing is caused by the inability to capture the very exact moment of the arc ignition on this equipment without different kind of automated triggering arrangements. The first image with a visible arc captures the event at or more likely slightly after the rise in welding current and actual ignition. Similarly, the last image frame with a visible arc is captured just before the arc ends. However, due to high power source data capture rates, all this happens within a limit of the time window presented with thousands of several power source data points between the capture consecutive images. This allows adequate matching between the image captures and the windows of the power source data.

Although exact synchronization may not be fully achievable due to mentioned limitations, a reasonable level can be attained by finding the closest possible time on the power source time domain for each image. Aligning of both data modalities between the matched starting event and continuing forward on the power source timeline is shown in Fig 5. In all but the first trial run, the lack of native timestamps in the collected power source data prevents determination of missing timestamps or clock skews. However, in the initial trial run with the same equipment, the timestamps and existing image timestamps show a linear increment in time with no missing stamps detected.

V. Aggregating the Complete Dataset

The power source data is split into segments limited to the intervals between consecutive images and their capture times, forming correlated image and power source data snippets as data point pairs. These pairs represent a window of measured power in and around the captured image, allowing for more varied data usage, such as processing power source data in the frequency domain matched and correlated with the images. Dividing the images equally along the power source time domain places each image within the running length of approximately 2,668 power source current and voltage values experienced around the image capture time. Scaling and aligning the images to the power source time scale for each image left some residues, which were trimmed to maintain consistent number of timesteps per image. Finally, all split pieces of power source data captures were trimmed to the minimum common length of the sets.

In the combined dataset containing all test runs' data, the mean drift between matching the image clock domain to power source clock was 6.320 nanoseconds with standard deviation of 3.632 for the image timestamps. Further visual inspection of the dataset convinces us that the synchronization is at a satisfactory level for our intended future trials . Some welding test runs have

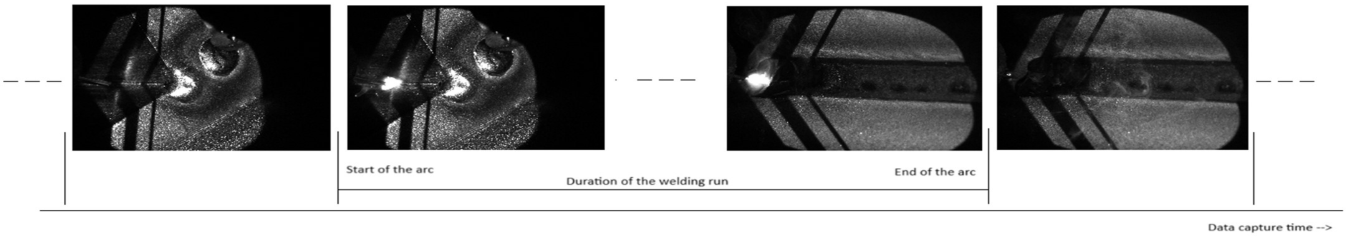

Fig. 4. Start and end of the welding arc as indicated visually in the sequence of image frames.

arbitrary interruptions during it which can be observed aligning well both in the power source and image data in time domain in comparable manner as identifying the start and stop points of the welding process itself.

All data combined with images' ROI and trimmed power source data produces a compressed collection with a file size of ~3.9 gigabytes. The subsets in the data are numbered from 1-16 after the test run numbering as presented in Table 3 in the appendix.

A. Power Source Time Series

In the raw data resulting from the test runs, the initial run provided correctly adjusted values for the welding voltage and the current. The consecutive runs used scaling values of -12.5 and -185.9608 for the voltage in volts and current in amperes, respectively. The power source time series in the final format will be 64-bit floating point values. Histogram of the value distribution in both current and voltage time series in complete dataset after scaling plotted on Fig 6. The distributions may not be ideal, e.g., for machine learning purposes without filtering or other manipulations.

On frequency domain after a standard Fast Fourier Transformation (FFT) a visible peak around 36kHz is noticed in the current signal as illustrated in Fig 7. This was explained to be caused by the welding equipment manufacturer by the welding equipment's own power source operating at 18kHz and its harmonics.

The Augmented Dicker-Fuller (ADF) [10] test for the complete power source time series shows highly negative ADF test statistics of -12.3613 and p-value of 0.0052 indicate convincing evidence against the null hypothesis of a unit root. The critical values for 1%, 5% and 10% were -3.4305, -2.8616 and -2.5668, respectively. According to the test results, the power source data time series fulfills the criteria for stationarity. Hence, the mean and variance remain constant over time and patterns remains stable over time. I.e., the mean and variance remain constant over time and patterns remains stable over time

B. Image Dataset

The collected grayscale image frames are ROI's separated from the original images with 8-bit data format yielding integer pixel values ranging from 0 to 255. Fig 8 plots a histogram of the pixel intensity value distribution on all ROI images in the dataset. The plot separates the subsets 1 and 2-16 as the imaging parameters were different resulting different distribution of the pixel values. Obviously there is a peak in very high valued or bright pixels due to high intensity signal emitted by the high temperature welding melt pool. This helps in identifying automatically the active welding arc and thus annotating images and the associated time series data with a binary flag.

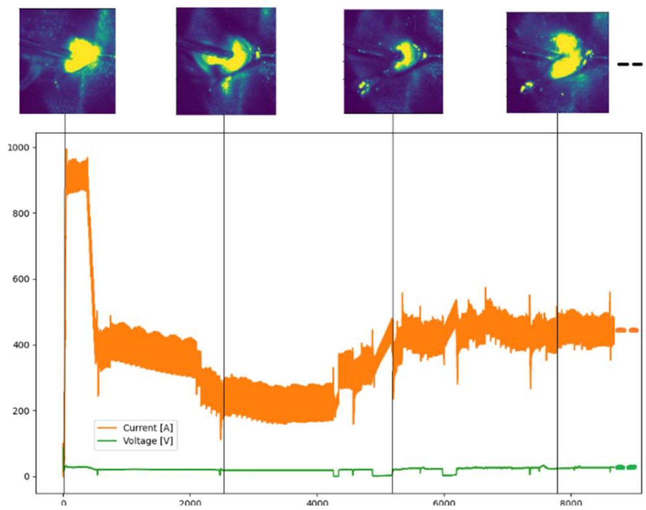

Fig. 5. Sample of image capture frames aligned to the electrical power source current and voltage voltage time domains, Scales are absolute captured values on the vertical axis and numbered time steps on horizontal zeroed to the arc ignition event.

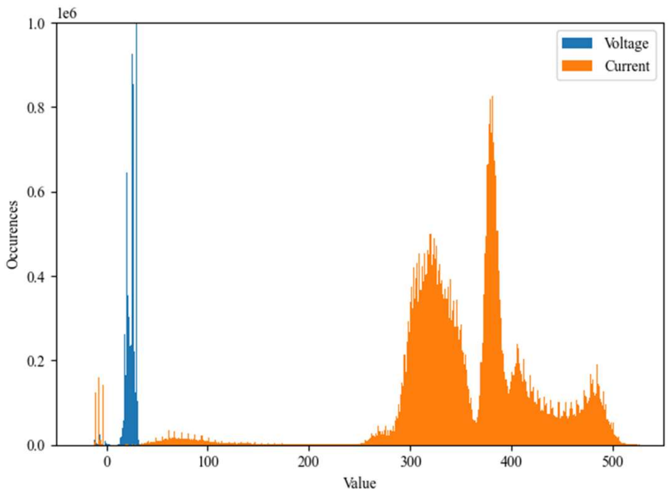

Fig. 6. Histogram of the value distribution in the current and voltage time series limited to current values under 550 amperes for eliminating nearly empty space.

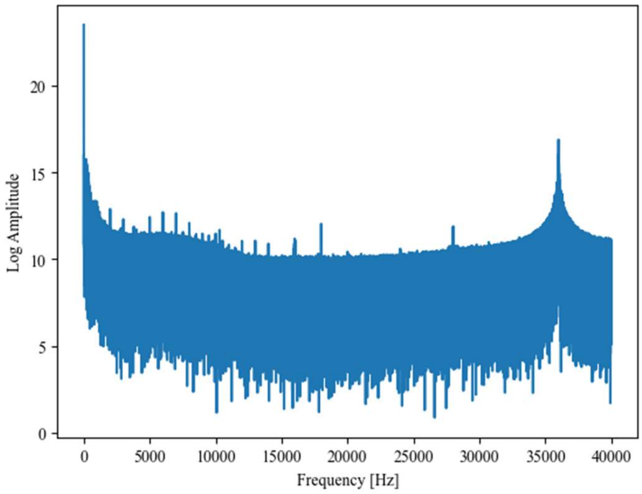

Fig. 7. Current power source data in frequency domain after FFT with a noticeable equipment sourced peak at 36kHz. For clarity, amplitude is logarithmic.

Based on the pixel value distribution, the ROI images were processed by iterating through each image and performing intensity analysis to identify welding activity. A threshold was established based on the intensity values, with subset 1 requiring a minimum frequency of 30 intensity values, while subsets 2-16 adhered to a threshold of 190. Once the thresholds were set, every image was reevaluated against the relevant defined parameter to determine its classification. Images that met or exceeded the threshold were labeled under heading "arc_on" with a binary designation of [0, 1], signifying detected welding activity. The structured approach ensured consistent classification across the dataset. An example set of ROI images with intensity histograms are presented in Fig 9. Also, the labeling of detected welding activity is mentioned and the frequency of detected high intensity pixels (255).

Selected samples of the finalized data are presented on the dataset repository introduction [1] due to space imitations here. The images are in the native captured size without ROI framing.

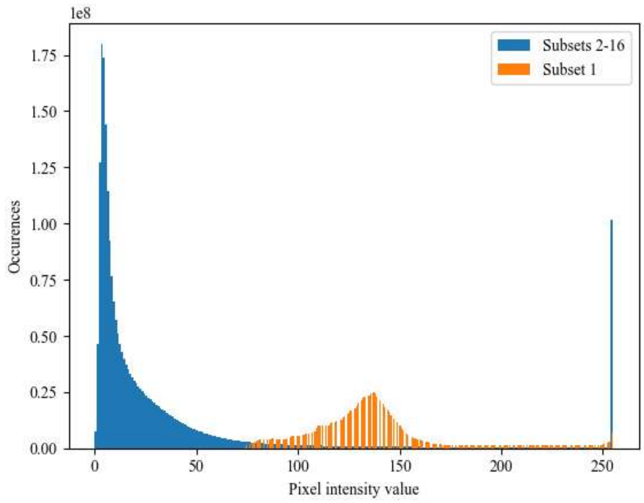

Fig. 8. Histogram of the pixel intensity value distribution, 0-255, on all ROI images in the dataset. Separately are the subsets 1 and 2-16 with different imaging parameters and thus different distribution of the pixel values.

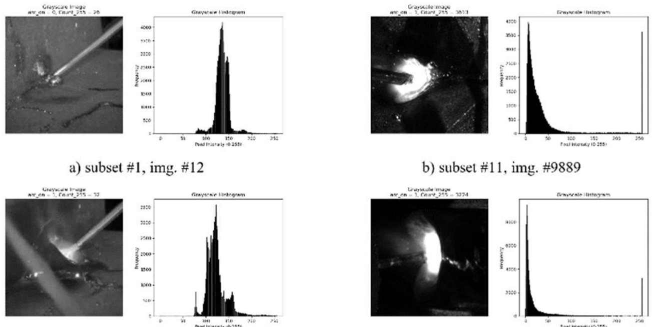

Fig. 9. ROI images with subsequent intensity histograms (A-D) for performing detection of welding activity "arc_on" [0, 1].

The power source data including voltage and current are from the active welding time. Some test runs experienced interruptions which are evidenced in the power source data.

VI. Discussion

The alignment was based on the identification of the active welding visually in the images and locating rising and falling edges of the voltage and power in the power source data. Even with this rudimentary method we achieved good results of presenting satisfyingly low levels of timing errors of +0.006320 milliseconds as timing drift between the images and power source data timing. Further, the Augmented Dicker-Fuller method indicated very acceptable level of stationarity in the time series data suggesting usefulness for machine and deep-learning purposes.

The additional labelling of the active welding is one of the many possible ways of enriching the dataset furthermore for supervised learning purposes. Some other potential new labels can be anticipated in a form of changing size and shape of the arc due to various shape features in the steel structure. These include e.g. general welding quality labels by the experts, crossing of the pre-welded tacks and varying size and shape of the airgap between the steel plates.

The multiple modalities here enable several potential usages with all or just selected modalities. Various multimodal models, like Variational Autoencoders (VAE) [11] could be used for generative and exploratory purposes, or e.g. anomaly detection in the latent space, in unsupervised learning manner. The initial trial with a specially constructed multimodal VAE, as in Fig 10, shows abilities for both anomaly detection and cross-modal generative purposes with input data. What is already anticipated is that the VAE and especially its encoder with fast inference speeds could provide anomaly detection for real time welding process control systems, in which diffusion models may lack the required processing speed.

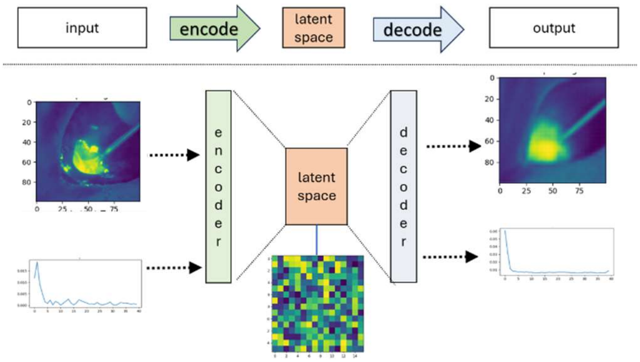

Fig. 10. Conceptual structure and flow of input and output data in a Variational Autoencoder system.

REFERENCES

[1] T. Aihkisalo, P. Tikka, A. Kahri, A. Liski, and K. Salmela, “KemppiPowerImages,” 2025, *Kaggle*. doi: 10.34740/KAGGLE/DS/7374740.

[2] Leonard P. Connor, Ed., *Welding Handbook*, vol. 1. American Welding Society, 1987. doi: 10.1007/978-1-349-10624-0.

[11] D. P. Kingma and M. Welling, “Auto-Encoding Variational Bayes,” Dec. 2013.

APPENDIX

TABLE I. WELDING PROCEDURE SPECIFICATIONS (WPS) ON EACH TEST RUN

WPS#	Material	Joint type	Welding Position	Material Thickness [mm]	Wire feed speed [m/min]	Travel speed [mm/min]	Stick out length [mm]	Gas flow [l/min]	Filler diam [mm]	Welding Current [A]	Welding Voltage [V]	Throat thickness[mm]
1	S355MC	FW	PB	15.0	10.5	400	22.0	18.0	1.2	342	29.8	5.0
2	S355MC	FW	PB	15.0	10.5	290	20.0	18.0	1.2	342	29.8	6.0
3	S355MC	FW	PF	15.0	3.3	210	15.0	18.0	1.2	123	18.2	6.0

TABLE II. WELDING TEST RUNS, APPLIED WELDING PROCEDURE SPECIFICATIONS AND WELDING STRUCTURE AND PROCEDURE SPECIFIC NOTES

Welding Run #	Applied WPS	Notes
1-3	1	-
4-5	1	Intentional stop and restart of welding
6-7	1	Two tacks of 50 mm and two tacks of 10 mm on welding path
8-9	1	Increasing air gap between plates from 0 to 6 mm
10-11	1	Tilted plate 0 to 3 mm
12-13	1	Four tacks of 50 mm
14-15	2	-
16	3	MAX Position (pulse MAG and short arc variating) straight travel

TABLE III. DATA DIMENSIONS OF THE COLLECTED DATA AND THE FINAL DIMENSIONS AFTER ALIGNMENT AND SYNCHRONIZATION.

Welding Run #	Images			Power source data points			
	Frames [#]	*FPS [f/sec]*	*Resolution [px]*	*Voltage [#]*	*Current [#]*	*Sampling rate [kHz]*	*Time synchronized size*
1	10017	60	720 x 540	17144551	17144551	100	9889 frames and power source data points of 1280 to 80 kHz sampling rate
2	2148	30	1440 x 1080	5949899	5961646	80	1824 frames and power source data points of 2630 steps each
3	2162	--"--	--"--	6450669	6426824	--"--	1725 frames and power source data points of 2629 steps each
4	2290	--"--	--"--	6688266	6663594	--"--	1933 frames and power source data points of 2655 steps each
5	2204	--"--	--"--	7110120	7123943	--"--	1899 frames and power source data points of 2618 steps each
6	2365	--"--	--"--	6807061	6780282	--"--	2095 frames and power source data points of 2554 steps each
7	2372	--"--	--"--	7434397	7404360	--"--	2093 frames and power source data

							points of 2504 steps each
8	1960	--“--	--“--	7767118	7781646	--“--	1639 frames and power source data points of 1625 steps each
9	2106	--“--	--“--	6221824	6233559	--“--	1723 frames and power source data points of 2047 steps each
10	2132	--“--	--“--	6577519	6590614	--“--	1854 frames and power source datapoints of 1727 steps each
11	1933	--“--	--“--	6471917	6484022	--“--	1694 frames and power source data points of 1565 steps each
12	2098	--“--	--“--	11877128	11835107	--“--	1880 frames and power source data points of 1657 steps each
13	2058	--“--	--“--	6446511	6459180	--“--	1859 frames and power source data points of 1825 steps each
14	491	--“--	--“--	2077285	2069325	--“--	343 frames and power source data points of 2332 steps each
15	385	--“--	--“--	4745662	4727716	--“--	282 frames and power source data points of 2152 steps each
16	1000	--“--	--“--	6422527	6433733	--“--	975 frames and power source data points of 2152 steps each
In total							33707 frames and power source data points of minimum 1280 timesteps